

Week 3 Interim Deliverable

Healthcare Workflows & Systems Thinking

Project Title: Caribbean ED Validation of Machine-Learning Triage Support

Student Name: Ansarah Mohammed

Draft Workflow Notes

For this interim deliverable, I reviewed the Mercer General ED workflow brief, the De Freitas et al. paper, and the “Understanding the Caribbean Healthcare ED Workflows” talk to map the existing triage and patient-flow process. The aim was to identify the main clinical steps, decision points, handoffs, delays, and possible areas where an AI triage support tool could realistically fit into the current workflow.

The Mercer ED workflow begins with patient arrival, which may occur by ambulance, walk-in, private vehicle, taxi, or EMS. The first formal step is registration, where demographic and administrative information is captured. After registration, the patient moves to triage, where the triage nurse records the chief complaint, vital signs, pain score, allergies, brief clinical history, and assigns an Emergency Severity Index (ESI) level. This triage decision determines the patient’s urgency and guides their movement into the appropriate care area.

The current workflow can be summarised as:

Patient arrival → Registration → Vital signs → Triage level assignment → Care area assignment → ED assessment → Investigations/treatment → Disposition → Exit

At Mercer, triage is mainly nurse-led. ESI 1 patients are directed to resuscitation for immediate intervention, ESI 2 and 3 patients are usually placed in acute care, and ESI 4 and 5 patients may be directed to fast-track or the waiting room. This makes the triage stage a key decision point because it affects patient prioritisation, bed allocation, staff workload, and downstream ED flow.

A major point from the workflow review is that triage is not a one-time event. Patients may deteriorate while waiting, especially during high-volume periods. Re-triage and waiting-room

reassessment are therefore important parts of the process. Any AI tool that only supports the first ESI assignment but does not account for reassessment could miss an important part of the real clinical workflow.

The De Freitas paper and workflow talk also highlighted that Caribbean EDs operate as complex systems rather than simple linear pathways. Patient flow is affected by staffing levels, paper-first documentation, delays in digital data entry, physical layout, handovers, diagnostic turnaround times, availability of ward beds, and dependence on other hospital departments. These factors are important because an AI model may perform well technically but still fail if it does not fit the operational environment.

For this project, the most realistic AI plug-in points appear to be around the vital signs, triage decision, and re-triage stages. At these points, the system could use information that is already available during triage, such as arrival mode, age, sex, chief complaint, vital signs, pain score, brief history, and initial ESI level. The tool should not depend on laboratory or imaging results during initial triage because those data are not available at that stage of the workflow.

The proposed AI support system should therefore be designed as a low-friction decision-support tool rather than a replacement for clinical judgement. A practical first step may be structured digital triage capture, which would create a de-identified dataset for audit, quality improvement, and future AI validation. This would support the long-term goal of developing a machine-learning triage support system while still respecting the time pressure and workflow constraints of a Caribbean emergency department.

Initial Workflow Diagram (In Git Hub):

The first-pass workflow diagram was created in Mermaid and uploaded to the project repository.

The diagram can be found in:

GitHub: <https://github.com/Kari-2005/carisurg-portfolio>

Location: docs/Week3/Initial_Week3InterimWorkflow.md

Explanation of Workflow and Candidate AI Plug-in Points:

The figure presents a first-pass workflow diagram of the Mercer General Emergency Department, beginning with patient arrival and ending with patient disposition and exit from the department. The diagram was developed using the Mercer General ED workflow brief together with insights from the De Freitas et al. paper and the "Understanding the Caribbean Healthcare ED Workflows" talk. It illustrates the major stages of the patient journey, the key decision points, and potential areas where AI support could be integrated without disrupting the existing workflow.

The process begins with patient arrival and registration, followed by the collection of vital signs and assignment of an Emergency Severity Index (ESI) level. Based on the assigned triage category, patients are directed to the appropriate care area, including resuscitation, acute care, or the fast-track/waiting room. Patients who remain in the waiting room may require re-triage if their condition deteriorates or prolonged waiting times occur. Following physician assessment, patients proceed through investigations and treatment before a final disposition decision is made, resulting in admission, discharge, transfer, or exit from the department.

Three candidate AI plug-in points were identified from the workflow.

AI Plug-in Point 1: Risk Check Following Vital Sign Collection

The first proposed AI plug-in point occurs immediately after vital signs are recorded. At this stage, the system would use information already available during triage, including age, sex, chief complaint, arrival mode, vital signs, and pain score. The output would be a suggested risk level or acuity flag for review by the triage nurse. The final triage decision would remain under the responsibility of the clinician, with the AI acting only as a decision-support tool rather than replacing clinical judgement.

AI Plug-in Point 2: Waiting-Room Re-triage Monitoring

The second AI plug-in point focuses on patients who have been directed to the waiting room or fast-track area. The system could monitor factors such as waiting time, initial ESI level, and previously recorded observations to identify patients who may require reassessment. The output

would be an alert recommending repeat vital signs or urgent review. This functionality would support the existing re-triage process and help identify deteriorating patients earlier.

AI Plug-in Point 3: Flow Visibility and Bottleneck Identification

The third AI plug-in point occurs during investigations and treatment. At this stage, the system could monitor delays associated with laboratory investigations, imaging studies, specialist consultations, and bed availability. The output would be a simple indicator highlighting areas where patient flow is being delayed. This information could support charge nurses and physicians in prioritising resources and improving overall throughput within the emergency department.

Overall, the diagram represents an initial systems-level view of the emergency department workflow and highlights potential opportunities for AI support. The proposed plug-in points are intended to augment existing clinical processes while maintaining clinician oversight and minimising disruption to the current workflow.

Candidate Workflow Constraints

Constraint 1: Paper-First Documentation and Data Availability

Mechanism

Patient information and vital signs may initially be recorded on paper before being entered into electronic systems. This creates a delay between when information becomes available to clinicians and when it becomes available digitally.

Design Impact

An AI system that relies entirely on real-time electronic health record (EHR) data may not have access to the necessary information during the triage process. This could limit the usefulness of the tool at the point where decisions are being made.

Mitigation

The proposed system should use information already collected during triage and support a gradual transition towards structured digital data capture. Initial implementations should minimise dependence on fully digitised records.

Constraint 2: Limited Triage Nurse Time and Cognitive Workload

Mechanism

Triage nurses operate under significant time pressure, particularly during periods of high patient volume. Additional tasks, excessive alerts, or complex interfaces may increase cognitive workload and disrupt existing workflows.

Design Impact

If the AI system requires substantial additional data entry or produces too many alerts, clinicians may ignore the tool or avoid using it altogether. This would reduce adoption and limit any potential benefit.

Mitigation

The AI tool should function as low-friction decision support and rely primarily on information already collected during triage. Outputs should be concise, clinically relevant, and designed to support rather than replace clinician judgement.

Constraint 3: Patient Deterioration During Waiting and Re-triage

Mechanism

Patient conditions may change after the initial triage decision, particularly during periods of prolonged waiting. As a result, the patient's original ESI category may no longer accurately reflect their current clinical condition.

Design Impact

An AI model based solely on initial triage information may fail to recognise deterioration occurring later in the patient journey. This could reduce the effectiveness of the system and limit its clinical usefulness.

Mitigation

The proposed system should support repeat observations and generate prompts for reassessment when appropriate. Incorporating re-triage into the workflow would allow the system to complement existing clinical practices and better reflect the dynamic nature of emergency department operations.

Overall, these constraints highlight the importance of designing AI systems that fit existing emergency department workflows and support clinical decision-making without introducing unnecessary complexity or disruption.

